PH 3120 - Computational Physics Laboratory I

Introduction to data analysis with Python

20 May - 26 May exercises part 1

Q1

```
a= 15
b= 6

print(a+b)
print(a - b )
print ( a*b )
print( a/b )
print(a**b )
```

```
21
9
90
2.5
11390625
```

Q2

```
s=float(input('enter the height'))
time=((2*s)/9.81)**1/2
print(time)
```

```
enter the height 5
```

```
0.509683995922528
```

Q3

```
import numpy as np
lst=[0,30,45,60,90]
for i in range(1):
    a=np.sin(np.deg2rad(lst))
```

```
    print ("sin",lst, "=",a)
    b=np.cos(np.deg2rad(lst))
    print ("cos",lst, "=",b)
```

```
sin [0, 30, 45, 60, 90] = [0.         0.5        0.70710678 0.8660254
1.        ]
cos [0, 30, 45, 60, 90] = [1.00000000e+00 8.66025404e-01 7.07106781e-
01 5.00000000e-01
 6.12323400e-17]
```

```
import numpy as np
pi=3.14
lst=[0,pi/6,pi/4,pi/3,pi/2]
for i in range(1):
    a=np.sin(lst)
    print ("sin",lst, "=",a)
```

```
sin [0, 0.5233333333333333, 0.785, 1.0466666666666666, 1.57] = [0.
0.4997701  0.70682518 0.86575984 0.99999968]
```

Q4

```
arrx=np.array([3,-2,1])
arry=np.array([2,-4,-3])
c=np.linalg.norm(arrx)
print("magnitude of the a",c)
d=np.linalg.norm(arry)
print("magnitude of the b",d)
arr1=arrx+arry
print("a+b =",arr1)
arr2=arrx-arry
print("a-b =",arr2)
arr3=arrx*arry
print("a*b =",arr3)
arr4=arrx/arry
print("a/b =",arr4)

print(2*arrx)

ar=np.dot(arrx,arry)
cos=ar/(c*d)
print(cos)
q=np.arccos(cos)
print("Angle bitween two vectors in radius ",q)
cross=np.array([np.cross(arrx,arry)])
print(cross)
```

```
magnitude of the a 3.7416573867739413
magnitude of the b 5.385164807134504
```

```
a+b = [ 5 -6 -2]
a-b = [1 2 4]
a*b = [ 6  8 -3]
a/b = [ 1.5         0.5         -0.33333333]
[ 6 -4  2]
0.5459208336840117
Angle bitween two vectors in radius  0.9933085441248494
[[10 11 -8]]
```

Q5

```
a=np.array([0,1,-1])
b=np.array([0,-2,-2])
c=np.array([0,2,-3])

for i in range (3):
    f=(3*(a[i]**2*c[i])-(a[i]*b[i]**3)+5)
    print(f)

5
19
-12
```

Q6

```
n=4.5
c=1760
t=600
N=n*(np.exp(-t/c))
print('Remains of an original mass in kg',N)


Remains of an original mass in kg 3.20005598788659

k=1220/np.log(2)
print (k)
n=4.5
t=600
N=n*(np.exp(-t/k))
print('Remains of an original mass in kg',N)


1760.0879498845354
Remains of an original mass in kg 3.2001105009718582
```

Q7

```
pw='tharaka'
imax=3
a=str(input("Enter your password"))
i=1
while pw!=a:
    print("Your password is incorrect")
    a=str(input("Enter your password"))
    i=i+1
    if i>imax:
        print("You cann't attempet again")
        break
else:
    print('Your password is correct')
```

```
Enter your password tharaka

Your password is correct
```

Q8

```
num1 =0
num2 =1
tot = 0
print (num1)
print(num2)

for i in range (20):
    a = num1 + num2
    if a<500:
        tot = tot + a
        print(a)
        num1 = num2
        num2 = a
    else:
        break

print("Total Sum is ",tot)
```

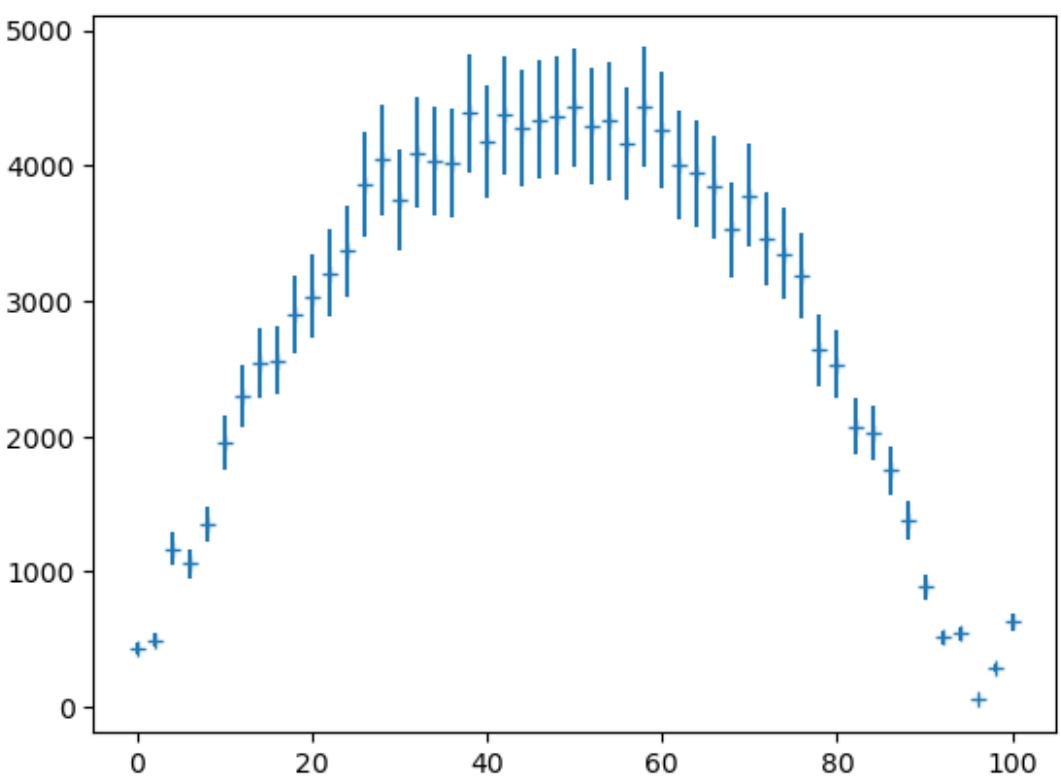
```
0
1
1
2
3
```

```
5
8
13
21
34
55
89
144
233
377
Total Sum is  985
```

Q9

```
import matplotlib.pyplot as plt
data=np.loadtxt('distance.dat')
x=data[:,0]
y=data[:,1]
err=data[:,2]

plt.errorbar(x,y,marker='+',yerr=err,ls='none')
plt.show()
```



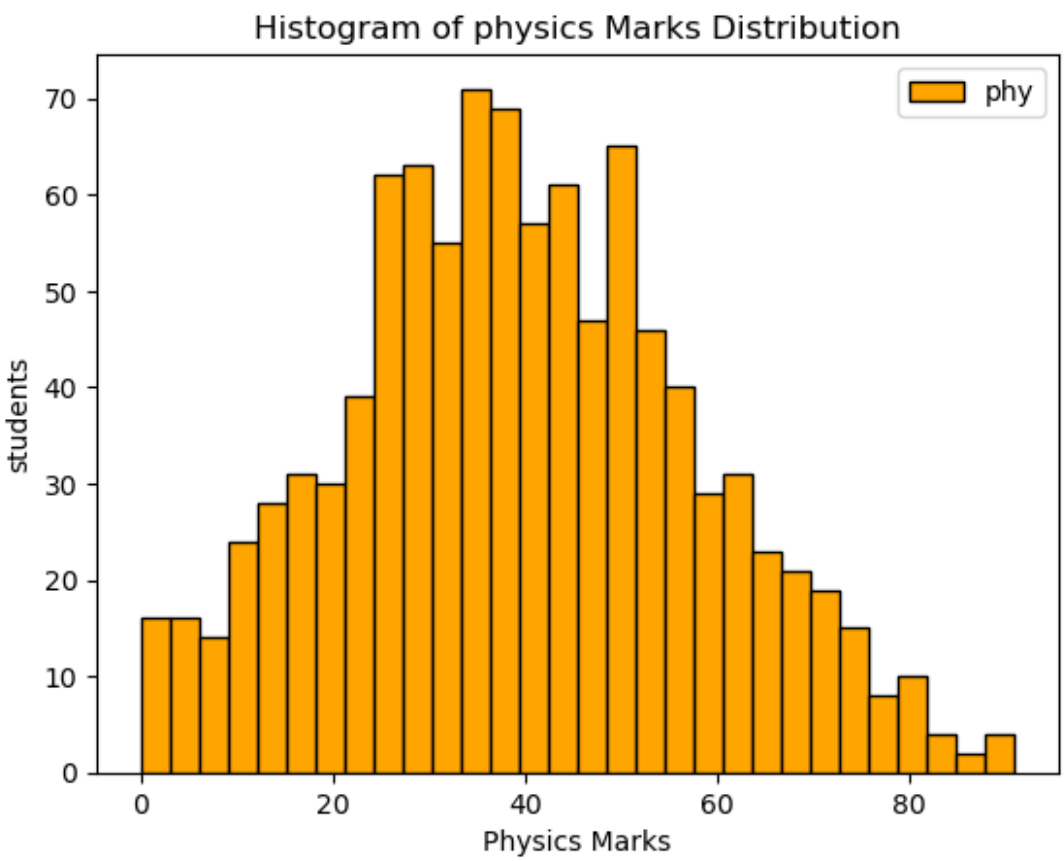
Q10

```
import numpy as np
import matplotlib.pyplot as plt
data = np.loadtxt('al-marks.dat')

phy = data[:,0]
chem = data[:,1]
math = data[:,2]
plt.hist(phy,color='orange',bins=30,edgecolor='black',label='phy')

plt.xlabel('Physics Marks')
plt.ylabel('students')
plt.title('Histogram of physics Marks Distribution')

plt.legend()
plt.show()
```



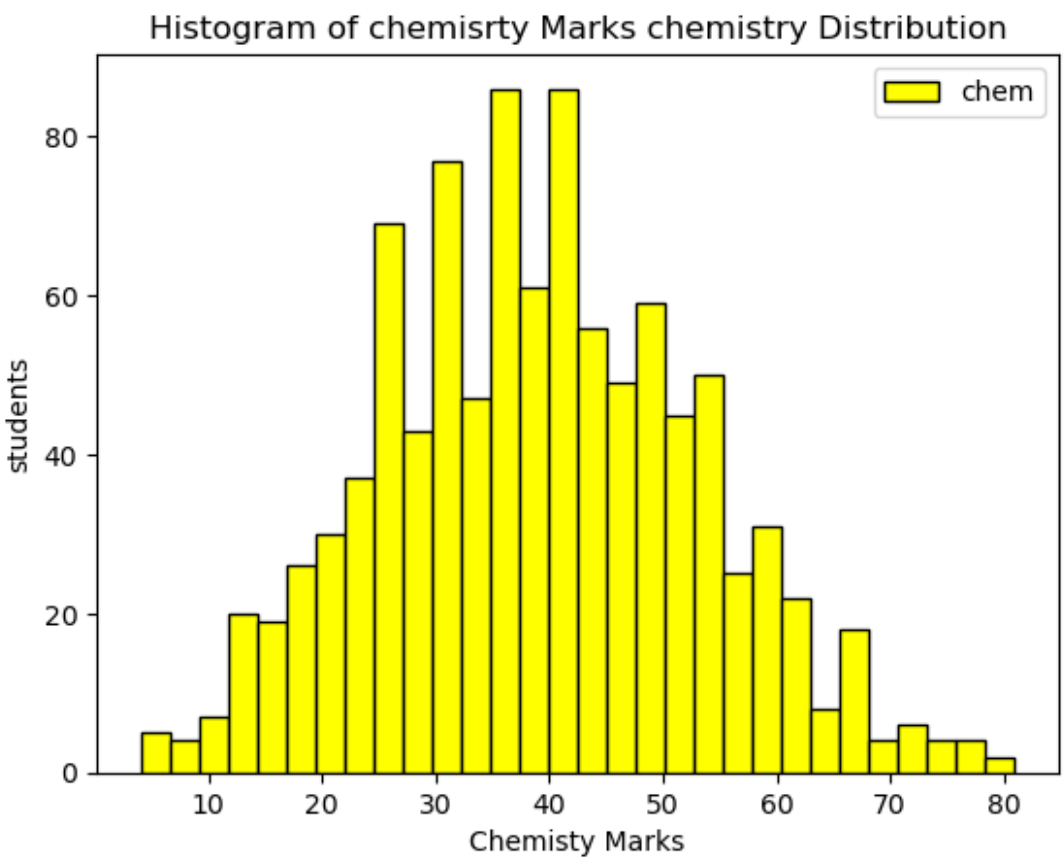

```
import matplotlib.pyplot as plt
data = np.loadtxt('al-marks.dat')

phy = data[:,0]
chem = data[:,1]
```

```
math = data[:,2]
plt.hist(chem,color='yellow',bins=30,edgecolor='black',label='chem')

plt.xlabel('Chemisty Marks')
plt.ylabel('students')
plt.title('Histogram of chemisrty Marks chemistry Distribution')

plt.legend()
plt.show()
```

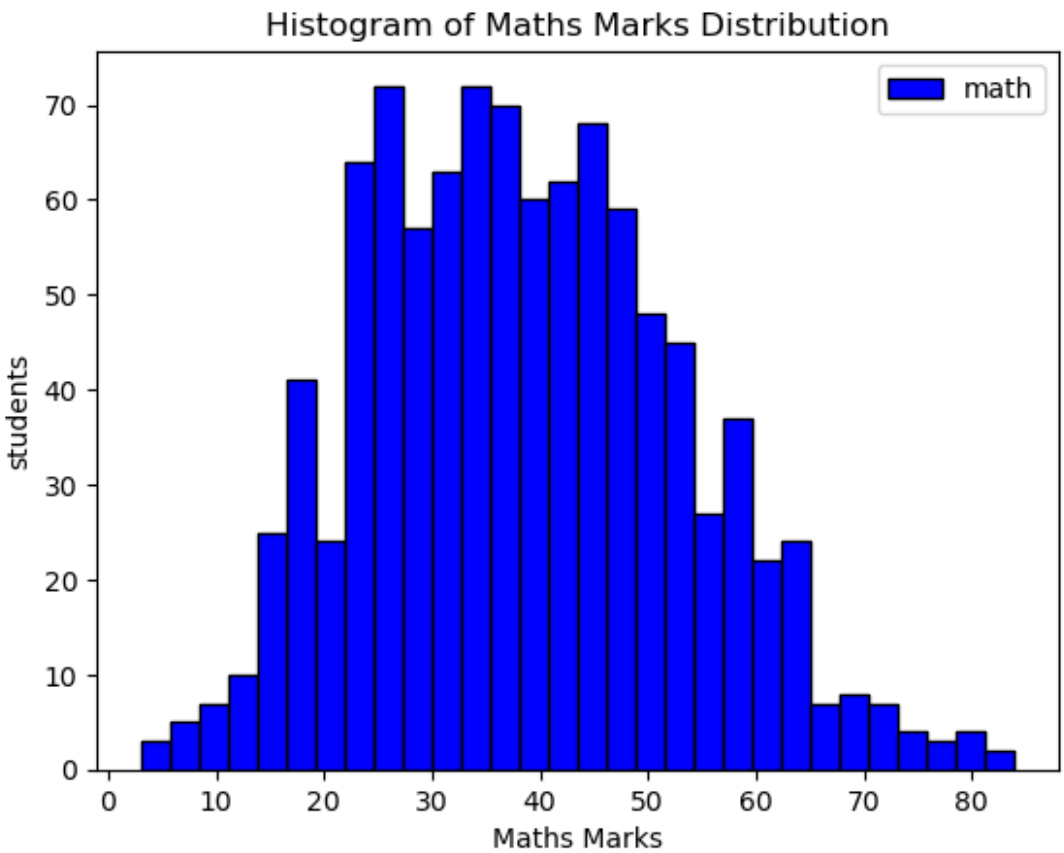



```
import matplotlib.pyplot as plt
data = np.loadtxt('al-marks.dat')

phy = data[:,0]
chem = data[:,1]
math = data[:,2]
plt.hist(math,color='blue',bins=30,edgecolor='black',label='math')

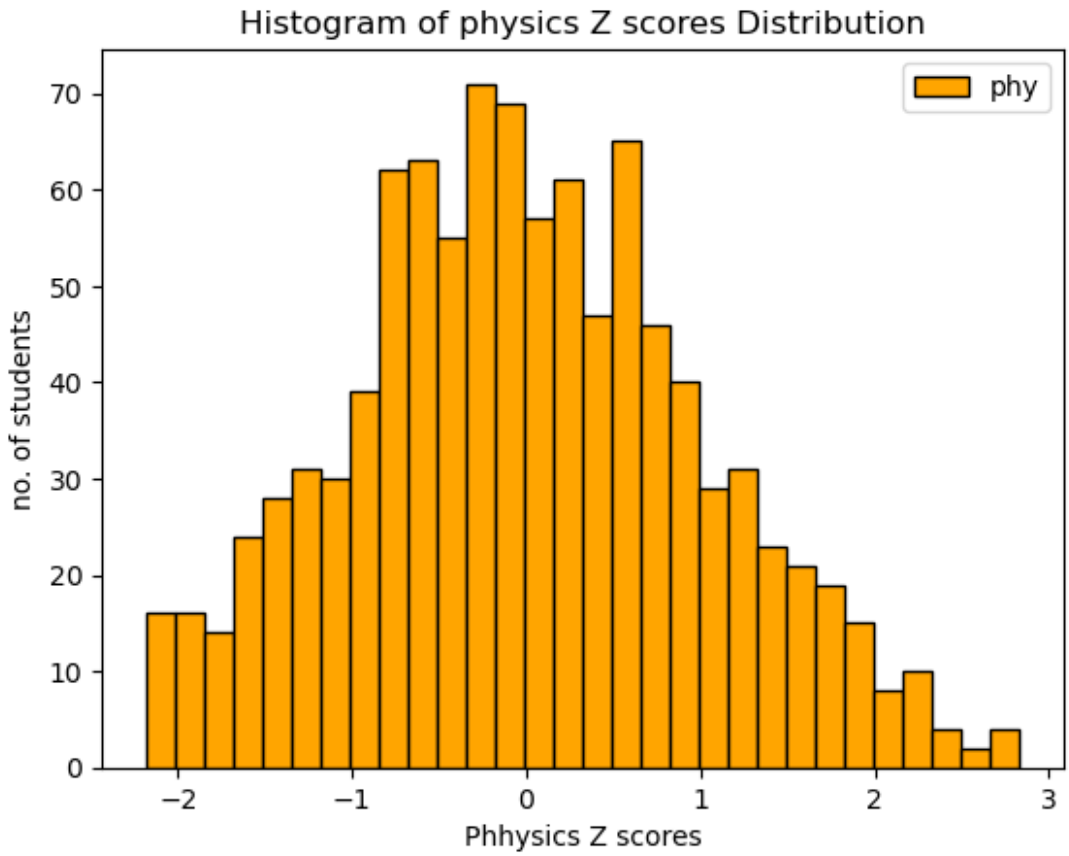
plt.xlabel('Maths Marks')
plt.ylabel('students')
plt.title('Histogram of Maths Marks Distribution')
```

```
plt.legend()
plt.show()
```

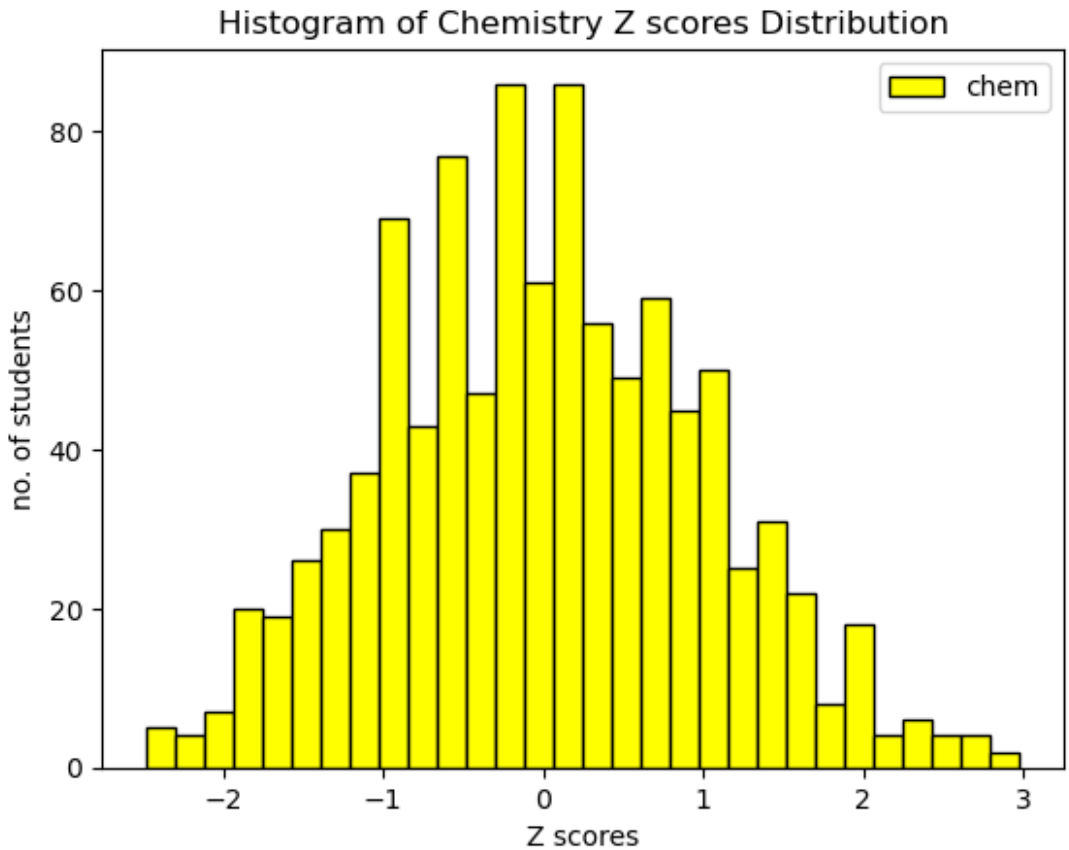



```
import matplotlib.pyplot as plt
data=np.loadtxt('al-marks.dat')
z=(phy-np.mean(phy))/np.std(phy)
plt.hist(z,color='orange',bins=30,edgecolor='black',label='phy')
plt.xlabel('Phhysics Z scores')
plt.ylabel('no. of students')
plt.title('Histogram of physics Z scores Distribution')
plt.legend()
plt.show

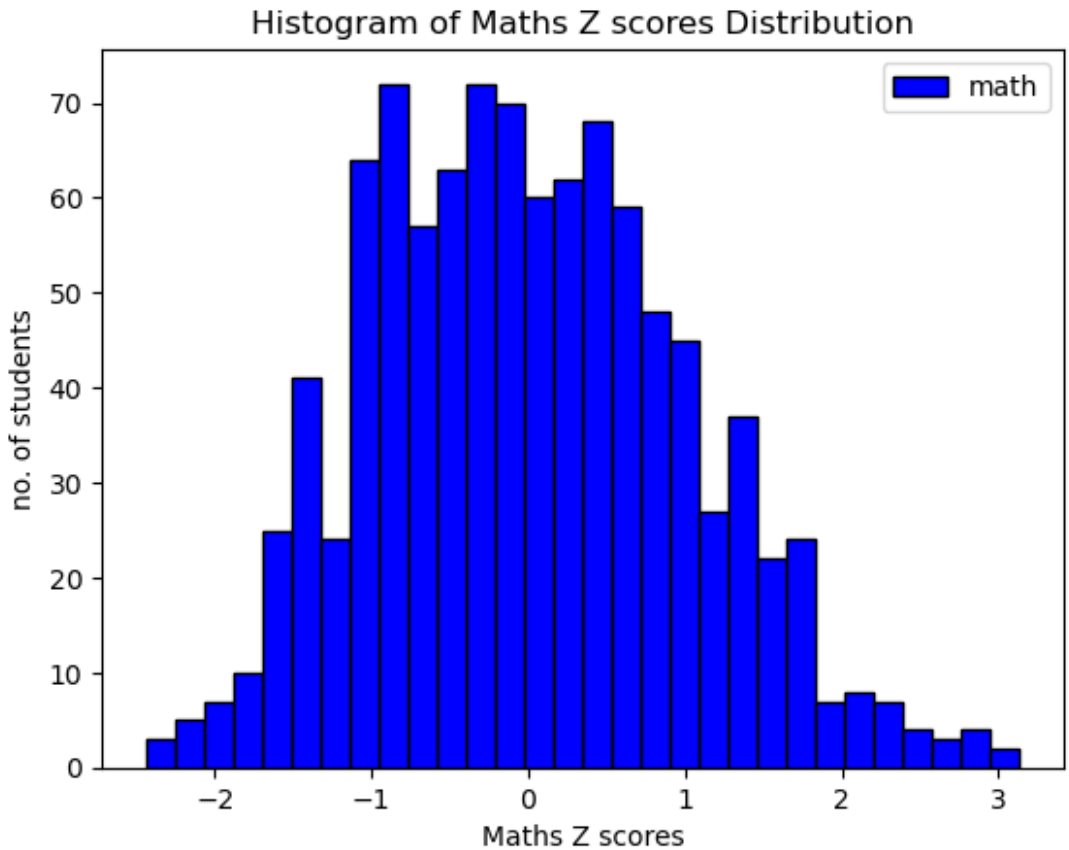
<function matplotlib.pyplot.show(close=None, block=None)>
```




```
data=np.loadtxt('al-marks.dat')
z=(chem-np.mean(chem))/np.std(chem)
plt.hist(z,color='yellow',bins=30,edgecolor='black',label='chem')
plt.xlabel('Chemistry Z scores')
plt.ylabel('no. of students')
plt.title('Histogram of Chemistry Z scores Distribution')
plt.legend()
plt.show()
```




```
data=np.loadtxt('al-marks.dat')
z=(math-np.mean(math))/np.std(math)
plt.hist(z,color='blue',bins=30,edgecolor='black',label='math')
plt.xlabel('Maths Z scores')
plt.ylabel('no. of students')
plt.title('Histogram of Maths Z scores Distribution')
plt.legend()
plt.show()
```




```
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